

Review

60%

26

Out of 43 points

2:33

Avg Time Per Question

Your Answers:

1 Multiple Choice 1 / 1 point

The payment of a 10% stock dividend by a company will result in an increase in that company's:

- ☐ A. current ratio.
- ☐ B. financial leverage.



☒ C. contributed capital.

Feedback

C is correct. A stock dividend is accounted for as a transfer of retained earnings to contributed capital.

2 Multiple Choice 1 / 1 point

If a company's common shares trade at very low prices, that company would be *most likely* to consider the use of a:

- ☐ A. stock split.
- ☐ B. stock dividend.



☒ C. reverse stock split.

Feedback

C is correct. A reverse stock split would increase the price per share of the stock to a higher, more marketable range that could possibly increase the number of investors who

would consider buying the stock.

3 Multiple Choice 1 / 1 point

In a recent presentation, Doug Pearce made two statements about dividends:

Statement 1:

"A stock dividend will increase share price on the ex-dividend date, all other things being equal."

Statement 2:

"One practical concern with a stock split is that it will reduce the company's price-to-earnings ratio."

Are Pearce's two statements about the effects of the stock dividend and stock split correct?



☒ A. No for both statements.

☐ B. Yes for Statement 1, and no for Statement 2.

☐ C. No for Statement 1, and yes for Statement 2.

Feedback

A is correct. Both statements are incorrect. A stock dividend will decrease the price per share, all other things being equal. A stock split will reduce the price and earnings per share proportionately, leaving the price-to-earnings ratio the same.

4 Multiple Choice 0 / 1 point

All other things being equal, the payment of an internally financed cash dividend is *most likely* to result in:

☐ A. a lower current ratio.

☐ B. a higher current ratio.



☐ C. the same current ratio.

Correct
Answer:

A. a lower current ratio.

Feedback

A is correct. By reducing corporate cash, a cash dividend reduces the current ratio, whereas a stock dividend (whatever the size) has no effect on the current ratio.

Vignette

John Ladan is an analyst in the research department of an international securities firm. Ladan is currently analyzing Yeta Products, a publicly traded global consumer goods company located in the United States. Selected data for Yeta are presented in Exhibit 1.

Exhibit 1 Selected Financial Data for Yeta Products

Most Recent Fiscal Year		Current	
Pretax income	US\$280 million	Shares outstanding	100 million
Net income after tax	US\$182 million	Book value per share	US\$25.60
Cash flow from operations	US\$235 million	Share price	US\$20.00
Capital expenditures	US\$175 million		
Earnings per share	US\$1.82		

Yeta currently does not pay a dividend, and the company operates with a target capital structure of 40% debt and 60% equity. However, on a recent conference call, Yeta's management indicated that they are considering four payout proposals:

Proposal #1: Issue a 10% stock dividend.

Proposal #2: Repurchase US\$40 million in shares using surplus cash.

Proposal #3: Repurchase US\$40 million in shares by borrowing US\$40 million at an after-tax cost of borrowing of 8.50%.

Proposal #4: Initiate a regular cash dividend.

5 Multiple Choice 1 / 1 point

The implementation of Proposal #1 would generally lead to shareholders:

☐ A. having to pay tax on the dividend received.

☐ B. experiencing a decrease in the total cost basis of their shares.



☒ C. having the same proportionate ownership as before implementation.

Feedback

C is correct. The implementation of Proposal #1, a stock dividend, would not affect a shareholder's proportionate ownership because all shareholders would receive the same proportionate increase in shares. Stock dividends, which are generally not taxable to shareholders, do not impact an investor's total cost basis (they merely reduce the cost basis per share).

A is incorrect because stock dividends are generally not taxable to shareholders. A stock dividend merely divides the "pie" (the market value of shareholders' equity) into smaller pieces.

B is incorrect because an investor's total cost basis will not be affected by a stock dividend; a stock dividend merely reduces the cost basis per share.

6 Multiple Choice 1 / 1 point

If Yeta's management implemented Proposal #2 at the current share price shown in Exhibit 1, Yeta's book value per share after implementation would be *closest* to:

☐ A. US\$25.20.



☒ B. US\$25.71.

☐ C. US\$26.12.

Feedback

B is correct. If Yeta implemented Proposal #2, a repurchase of US\$40 million in shares, the resulting book value per share (BVPS) would be US\$25.71, calculated as follows:

1. Yeta has a current BVPS of US\$25.60; therefore, total book value of equity is US\$2,560 million (= US\$25.60 × 100,000,000 shares).
2. The number of shares Yeta would repurchase is US\$40 million/US\$20.00 per share = 2 million shares.
3. Yeta shareholders' book value of equity after the buyback would be US\$2,520 million (= US\$2,560 million – US\$40 million).
4. The number of shares after the buyback would be 98 million (= 100 million – 2 million).
5. The BVPS after the buyback would be US\$2,520 million/98 million = US\$25.71.

A is incorrect because US\$25.20 incorrectly uses 100 million shares instead of 98 million shares in calculating BVPS after the buyback: $\text{US\$2,520 million} / 100 \text{ million} = \text{US\$25.20}$.

C is incorrect because US\$26.12 incorrectly uses US\$2,560 million (current book value) instead of US\$2,520 million as the book value of equity in calculating BVPS after the buyback. The BVPS after the buyback is incorrectly calculated as $\text{US\$2,560 million} / 98 \text{ million} = \text{US\$26.12}$.

7 Multiple Choice 0 / 1 point

Based on Exhibit 1, if Yeta's management implemented Proposal #3 at the current share price, earnings per share would:

☒ A. decrease.

Correct Answer:

C. increase.

☐ B. remain unchanged.

☐ C. increase.

Feedback

C is correct. In the case of external funding, a company's earnings per share will increase if the stock's earnings yield, which is the ratio of earnings per share to share price, exceeds the after-tax cost of borrowing. Yeta's earnings yield is 9.10% ($= \text{US\$1.82} / \text{US\$20.00}$), which exceeds the after-tax cost of borrowing of 8.50%.

A is incorrect because EPS will increase (not decrease) if the stock's earnings yield ($= \text{US\$1.82} / \text{US\$20.00}$) exceeds the after-tax cost of borrowing. Yeta's earnings yield of 9.10% exceeds the after-tax cost of borrowing of 8.50%.

B is incorrect because EPS will increase (not remain unchanged) if the stock's earnings yield ($= \text{US\$1.82} / \text{US\$20.00}$) exceeds the after-tax cost of borrowing. Yeta's earnings yield of 9.10% exceeds the after-tax cost of borrowing of 8.50%.

8 Multiple Choice 1 / 1 point

Based on Yeta's target capital structure, Proposal #4 will *most likely*:

☒ A. increase the default risk of Yeta's debt.

☐ B. increase the agency conflict between Yeta's shareholders and managers.

☐ C. decrease the agency conflict between Yeta's shareholders and bondholders.

Feedback

A is correct. Yeta is financed by both debt and equity; therefore, paying dividends can increase the agency conflict between shareholders and bondholders. The payment of dividends reduces the cash cushion available for the disbursement of fixed required payments to bondholders. All else equal, dividends increase the default risk of debt.

B is incorrect because the agency conflict between shareholders and managers would decrease (not increase) with the payment of dividends. Paying out free cash flow to equity in dividends would constrain managers in their ability to overinvest by taking on negative net present value (NPV) projects.

C is incorrect because paying dividends can increase (not decrease) the agency conflict between shareholders and bondholders. The payment of dividends would reduce the cash cushion available to Yeta for the disbursement of fixed required payments to bondholders. The payment of dividends transfers wealth from bondholders to shareholders and increases the default risk of debt.

9 Multiple Choice 1 / 1 point

The implementation of Proposal #4 would *most likely* signal to Ladan and other investors that future earnings growth can be expected to:

- ☐ A. decrease.
- ☐ B. remain unchanged.

☒ C. increase.

Feedback

C is correct. Dividend initiations convey positive information and are associated with future earnings growth, whereas dividend omissions or reductions convey negative information and are associated with future earnings problems.

A is incorrect because dividend initiations convey positive information and are associated with an expected increase (not a decrease) in future earnings growth. Dividend omissions or reductions convey negative information and are associated with future earnings problems.

B is incorrect because dividend initiations convey positive information and are associated with an expectation that future earnings growth will increase (not remain unchanged). In contrast, dividend omissions or reductions convey negative information and are associated with future earnings problems.

10 Essay 1 / 1 point

Match the phrases in Column A with the corresponding dividend theory in Column B. Note that you may use the answers in Column B more than once.

Column A

1. Bird in the hand
2. Homemade dividends
3. High tax rates on dividends

Column B

- a) Dividend policy matters
- b) Dividend policy is irrelevant



1 -> a
2 -> b
3 -> a

Feedback

The appropriate matches are as follows:

Column A

1. Bird in the hand
2. Homemade dividends
3. High tax rates on dividends

Column B

- a) Dividend policy matters
- b) Dividend policy is irrelevant
- a) Dividend policy matters

11 Multiple Choice 1 / 1 point

Which of the following assumptions is *not* required for Miller and Modigliani's (MM) dividend theory?

- ☐ A. Shareholders have no transaction costs when buying and selling shares.
- ☐ B. There are no taxes.



☒ C. Investors prefer dividends over uncertain capital gains.

Feedback

C is correct. The MM dividend theory assumes no taxes or transaction costs, but it does not assume investors have a preference for dividends over capital gains.

12 Multiple Choice 0 / 1 point

Sophie Chan owns 100,000 shares of PAT Company. PAT is selling for €40 per share, so Chan's investment is worth €4,000,000. Chan reinvests the gross amount of all dividends received to purchase additional shares. Assume that the clientele for PAT shares consists of tax-exempt investors. If PAT pays a €1.50 dividend, Chan's new share ownership after reinvesting dividends at the ex-dividend price is *most* likely to be closest to:

☐ A. 103,600.

☒ B. 103,750.

Correct Answer: **C. 103,900.**

☐ C. 103,900.

Feedback

C is correct. Because the clientele for PAT investors has the same tax rate (zero) for dividends and capital gains, the ex-dividend stock price of PAT should decline by the amount of the dividend to $€40 - €1.50 = €38.50$. Chan will purchase $€150,000 / €38.50 = 3,896$ additional shares. This increases her total shares owned to 103,896. Chan's new share ownership is closest to 103,900.

13 Multiple Choice 0 / 1 point

Which of the following is *most* likely to signal negative information concerning a company?

☐ A. Share repurchase.

☐ B. Decrease in the quarterly dividend rate.

☒ C. A two-for-one stock split.

Correct Answer: **B. Decrease in the quarterly dividend rate.**

Feedback

B is correct. A decrease in the quarterly dividend rate is likely to signal negative information. A decrease is typically understood as signaling poor future business prospects.

14 Multiple Choice 1 / 1 point

WL Corporation is located in a jurisdiction that has a 40% corporate tax rate on pretax income and a 30% personal tax rate on dividends. WL distributes all its after-tax income to shareholders. What is the effective tax rate on WL pretax income distributed in dividends?

☐ A. 42%.



☒ B.
58%.

☐ C. 70%.

Feedback

B is correct. The effective tax rate can be computed as 1 minus the fraction of 1 unit of earnings that investors retain after all taxes, or $1 - (1 - 0.40)(1 - 0.30) = 0.58$ or 58% effective tax rate. Another way to obtain the solution: Corporate taxes = $1.00 \times 0.40 = 0.40$ and Personal taxes = $0.60 \text{ in dividends} \times 0.30 = 0.18$, so Total tax = $0.40 + 0.18 = 0.58$, or 58% effective rate.

15 Multiple Choice 0 / 1 point

Which of the following factors is *least likely* to be associated with a company having a low dividend payout ratio?



☒ A. High flotation costs on new equity issues.

Correct
Answer:

C. Low growth prospects.

☐ B. High tax rates on dividends.

☐ C. Low growth prospects.

Feedback

C is correct. With low growth prospects, a company would typically have a high payout ratio, returning funds to its shareholders rather than retaining funds.

16 Multiple Choice 0 / 1 point

The dividend policy of Berkshire Gardens Inc. can be represented by a gradual adjustment to a target dividend payout ratio. Last year Berkshire had earnings per share of US\$3.00 and paid a dividend of US\$0.60 a share. This year it estimates earnings per share will be US\$4.00. Find its dividend per share for this year if it has a 25% target payout ratio and uses a five-year period to adjust its dividend.

☐ A. US\$0.68.



☒ B.
US\$0.80.

Correct
Answer:

A. US\$0.68.

☐ C. US\$0.85.

Feedback

A is correct. The estimated dividend per share is US\$0.68.

Previous DPS = US\$0.60

Expected EPS = US\$4

Target payout ratio = 0.25

Five-year adjustment factor = $1/5 = 0.2$

Expected dividend = Previous dividend + (Expected earnings $\times$ Target payout ratio – Previous dividend) $\times$ Adjustment factor
 = US\$0.60 + [(US\$4.00 $\times$ 0.25 – US\$0.60) $\times$ 0.2]
 = US\$0.60 + US\$0.08
 = US\$0.68

17 Multiple Choice 0 / 1 point

Beta Corporation is a manufacturer of inflatable furniture. Which of the following scenarios best reflects a stable dividend policy for Beta?

- ☐ A. Maintaining a constant dividend payout ratio of 40–50%.
- ☐ B. Maintaining the dividend at US\$1.00 a share for several years given no change in Beta's long-term prospects.



☒ C. Increasing the dividend 5% a year over several years to reflect the two years in which Beta recognized mark-to-market gains on derivatives positions.

Correct
Answer:

B. Maintaining the dividend at US\$1.00 a share for several years given no change in Beta's long-term prospects.

Feedback

B is correct. Choice A is consistent with a constant dividend target payout ratio policy. Choice C is not correct because the earnings increases described are not sustainable long term.

18 Multiple Choice 1 / 1 point

A company has 1 million shares outstanding and earnings are £2 million. The company decides to use £10 million in surplus cash to repurchase shares in the open market. The company's shares are trading at £50 per share. If the company uses the entire £10 million of surplus cash to repurchase shares at the market price, the company's earnings per share will be *closest* to:

- ☐ A. £2.00.
- ☐ B. £2.30.
- ☒ C. £2.50.

Feedback

C is correct. At the current market price, the company can repurchase 200,000 shares (£10 million/£50 = 200,000 shares). The company would have 800,000 shares outstanding after the repurchase (1 million shares – 200,000 shares = 800,000 shares).

EPS before the buyback is £2.00 (£2 million/1 million shares = £2.00). Total earnings after the buyback are the same because the company uses surplus (nonearning) cash to purchase the shares, but the number of shares outstanding is reduced to 800,000. EPS increases to £2.50 (£2 million/ 800,000 shares = £2.50).

19 Multiple Choice 0 / 1 point

Devon Ltd. common shares sell at US\$40 a share, and their estimated price-to-earnings ratio (P/E) is 32. If Devon borrows funds to repurchase shares at its after-tax cost of debt of 5%, its EPS is *most likely* to:

- ☐ A. increase.
- ☐ B. decrease.
- ☒ C. remain the same.

Correct Answer: **B. decrease.**

Feedback

B is correct. If the P/E is 32, the earnings-to-price ratio (earnings yield or E/P) is $1/32 = 3.125\%$. When the cost of capital is greater than the earnings yield, earnings dilution will result from the buyback.

20 Multiple Choice 1 / 1 point

A company can borrow funds at an after-tax cost of 4.5%. The company's stock price is US\$40 per share, earnings per share is US\$2.00, and the company has 15 million shares outstanding. If

the company borrows just enough to repurchase 2 million shares of stock at the prevailing market price, that company's earnings per share is *most likely* to:



☒ A.
increase.

☐ B. decrease.

☐ C. remain the same.

Feedback

A is correct. The company's earnings yield (E/P) is $\text{US\$2/US\$40} = 0.05$. When the earnings yield is greater than the after-tax cost of borrowed funds, EPS will increase if shares are repurchased using borrowed funds.

21 Multiple Choice 1 / 1 point

Crozet Corporation plans to borrow just enough money to repurchase 100,000 shares. The following information relates to the share repurchase:

Shares outstanding before buyback 3.1 million

Earnings per share before buyback US\$4.00

Share price at time of buyback US\$50

After-tax cost of borrowing 6%

Crozet's earnings per share after the buyback will be *closest* to:



☒ A.
US\$4.03.

☐ B. US\$4.10.

☐ C. US\$4.23.

Feedback

A is correct.

Total earnings before buyback: $\text{US\$4.00} \times 3,100,000 \text{ shares} = \text{US\$12,400,000}$

Total amount of borrowing: $\text{US\$50} \times 100,000 \text{ shares} = \text{US\$5,000,000}$

After-tax cost of borrowing the amount of funds needed: $\text{US\$5,000,000} \times 0.06 = \text{US\$300,000}$

Number of shares outstanding after buyback: $3,100,000 - 100,000 = 3,000,000$

EPS after buyback: $(\text{US\$}12,400,000 - \text{US\$}300,000)/3,000,000 \text{ shares} = \text{US\$}4.03$

The P/E before the buyback is $\text{US\$}50/\text{US\$}4 = 12.5$; thus, the E/P is 8%. The after-tax cost of debt is 6%; therefore, EPS will increase.

22 Multiple Choice 1 / 1 point

A company with 20 million shares outstanding decides to repurchase 2 million shares at the prevailing market price of €30 per share. At the time of the buyback, the company reports total assets of €850 million and total liabilities of €250 million. As a result of the buyback, that company's book value per share will *most likely*:

- ☐ A. increase.
- ☐ B. decrease.
- ☒ C. remain the same.

Feedback

C is correct. The company's book value before the buyback is €850 million in assets – €250 million in liabilities = €600 million. Book value per share is €600 million/20 million = €30 per share. The buyback will reduce equity by 2 million shares at the prevailing market price of €30 per share. The book value of equity will be reduced to €600 million – €60 million = €540 million, and the number of shares will be reduced to 18 million; €540 million/18 million = €30 book value per share. If the prevailing market price is equal to the book value per share at the time of the buyback, book value per share is unchanged.

23 Multiple Choice 1 / 1 point

An analyst gathered the following information about a company:

Number of shares outstanding	10 million
Earnings per share	US\$2.00
P/E	20
Book value per share	US\$30

If the company repurchases 1 million shares at the prevailing market price, the resulting book value per share will be *closest* to:

☐ A. US\$26.

☐ B. US\$27.



☒ C.
US\$29.

Feedback

C is correct. The prevailing market price is $\text{US\$}2.00(20) = \text{US\$}40.00$ per share; thus, the buyback would reduce equity by US\$40 million. Book value of equity before the buyback is US\$300 million. Book value of equity after the buyback would be $\text{US\$}300 \text{ million} - \text{US\$}40 \text{ million} = \text{US\$}260 \text{ million}$. The number of shares outstanding after the buyback would be 9 million. Thus, book value per share after the buyback would be $\text{US\$}260 \text{ million} / 9 \text{ million} = \text{US\$}28.89 \approx \text{US\$}29$.

24 Multiple Choice 0 / 1 point

If a company's objective is to support its stock price in the event of a market downturn, it would be advised to authorize:

☐ A. an open market share repurchase plan to be executed over the next five years.



☒ B. a tender offer share repurchase at a fixed price effective in 30 days.

Correct
Answer:

A. an open market share repurchase plan to be executed over the next five years.

☐ C. a Dutch auction tender offer effective in 30 days.

Feedback

A is correct. Of the three methods, only an authorized open market share repurchase plan allows the company the flexibility to time share repurchases to coincide with share price declines.

25 Multiple Choice 1 / 1 point

A company has positive free cash flow and is considering whether to use the entire amount of that free cash flow to pay a special cash dividend or to repurchase shares at the prevailing market price. Shareholders' wealth under the two options will be equivalent unless the:

☐ A. company's book value per share is less than the prevailing market price.

☐ B. company's book value per share is greater than the prevailing market price.



- ☒ C. tax consequences and/or information content for each alternative is different.

Feedback

C is correct. For the two options to be equivalent with respect to shareholders' wealth, the amount of cash distributed, the taxation, and the information content must be the same for both options.

26 Multiple Choice 0 / 1 point

Assume that a company is based in a country that has no taxes on dividends or capital gains. The company is considering either paying a special dividend or repurchasing its own shares. Shareholders of the company would have:

- ☐ A. greater wealth if the company paid a special cash dividend.



- ☒ B. greater wealth if the company repurchased its shares.

Correct
Answer:

C. the same wealth under either a cash dividend or share repurchase program.

- ☐ C. the same wealth under either a cash dividend or share repurchase program.

Feedback

C is correct. When there are no taxes, there are no tax differences between dividends and capital gains. All other things being equal, the effect on shareholder wealth of a dividend and a share repurchase should be the same.

27 Multiple Choice 1 / 1 point

Investors may prefer companies that repurchase their shares instead of paying a cash dividend when:



- ☒ A. capital gains are taxed at lower rates than dividends.

- ☐ B. capital gains are taxed at the same rate as dividends.
- ☐ C. the company needs more equity to finance capital expenditures.

Feedback

A is correct. When capital gains are taxed at lower rates than dividends, investors may prefer companies that return cash to shareholders through share repurchases rather than

dividends.

Vignette

Janet Wu is treasurer of Wilson Chemical Company, a manufacturer of specialty chemicals used in industrial manufacturing and increasingly in technology applications. Wilson Chemical is selling one of its older divisions for US\$70 million cash. Wu is considering whether to recommend a special dividend of US\$70 million or a repurchase of 2 million shares of Wilson common stock in the open market. She is reviewing some possible effects of the buyback with the company's financial analyst. Wilson has a long-term record of gradually increasing earnings and dividends.

28 Multiple Choice 1 / 1 point

Wilson's share buyback could be a signal that the company:

- ☐ A. is decreasing its financial leverage.
- ☒ B. views its shares as undervalued in the marketplace.
- ☐ C. has more investment opportunities than it could fund internally.

Feedback

B is correct. Management sometimes undertakes share repurchases when it views shares as being undervalued in the marketplace.

29 Multiple Choice 1 / 1 point

The most likely tax environment in which Wilson Chemical's shareholders would prefer that Wilson repurchase its shares (share buybacks) instead of paying dividends is one in which:

- ☐ A. the tax rate on capital gains and dividends is the same.
- ☐ B. capital gains tax rates are higher than dividend income tax rates.
- ☒ C. capital gains tax rates are lower than dividend income tax rates.

Feedback

C is correct. Shareholders would prefer that the company repurchase its shares instead of paying dividends when the tax rate on capital gains is lower than the tax rate on dividends.

Vignette

Barbara Carlyle is a financial adviser to high-net-worth individuals. She is currently reviewing the equity portfolio of a client and is considering adding new securities to it, as the client has indicated a preference for more income-producing securities. With this in mind, Carlyle takes a closer look at Avignon Corporation ("Avignon"), a chain of Canadian boutiques that has recently registered unusually high sales, resulting in large increases in the company's cash balance. Avignon's current stock price is C\$47.33. Exhibit 1 shows earnings and dividends for the preceding four years.

Exhibit 1 Avignon Corporation Earnings and Dividend History

	2013	2014	2015	2016
EPS	C\$1.38	C\$1.39	C\$1.37	C\$1.44
DPS	C\$0.20	C\$0.22	C\$0.22	C\$0.22

Carlyle reviews analysts' reports. She notes the significant change in cash due to the high sales volume and wonders whether that will prompt a dividend increase. However, most analysts have stated that because the industry is cyclical, the increase in sales is believed to be temporary.

Carlyle asks her assistant, Richard Lee, to investigate whether Avignon might use its surplus cash for a share repurchase rather than for dividends. Lee, a junior analyst, comments that share repurchases can be beneficial for several reasons:

1. The distribution of cash among shareholders is equivalent to what would have otherwise been distributed to them as dividends.
2. Share repurchases provide greater flexibility to management than the payment of cash dividends.
3. When directly negotiated, share repurchases can be used to purchase stock for less than the current market price.

Lee believes that looking at other companies that have completed share repurchases could be helpful to his analysis. He looks at the history of SpeedyPro Inc. ("SpeedyPro"), a US-based industrial services company whose business depends heavily on the petroleum exploration and production sector. SpeedyPro made its first share repurchase in early 2017 using surplus cash. SpeedyPro's selected financial information just prior to the repurchase is shown in Exhibit 2.

Exhibit 2 SpeedyPro, Inc. Selected Financial Information as of Year-End 2016

Net income	\$124 million
EPS	\$1.24
Shares outstanding	100 million
Details of shares repurchase	
Cash available for repurchase	\$836 million
Share price at the time of repurchase	\$38.00

Premium over current share price for repurchase

10.0%

Lee returns to Carlyle to continue the discussion. Carlyle explains to Lee that a complete analysis of the impact of a share repurchase should also include an evaluation of the effects on leverage. She points out that Avignon's most recent bond issue includes a covenant that limits the company's debt-to-equity ratio to 35%. She asks Lee to prepare an analysis for Avignon, using the information in Exhibit 3, to see if the debt covenant will be violated if the company repurchases shares.

Exhibit 3 Avignon Corporation Selected Financial Information as of Year-End 2016

Book value of equity	C\$3,600 million
Shares outstanding	200 million
Cash available for repurchase	C\$155 million
Debt-to-equity ratio	30.0%
After-tax cost of debt	5.0%

30 Multiple Choice 0 / 1 point

Based on Exhibit 1, Avignon's current dividend policy is *best* described as a:

☒ stable dividend policy.

Correct Answer Feedback: Correct. Avignon's current dividend policy would be classified as stable because it has paid the same amount for the past three years regardless of earnings volatility.

☐ residual dividend policy.

Incorrect Answer Feedback: Incorrect. A residual dividend policy is based on paying out the full amount of internally generated funds after capital expenditures. It is rarely used in practice because it typically results in highly volatile dividend payments.

☐ constant dividend payout ratio policy.

Incorrect Answer Feedback: Incorrect. A constant dividend payout ratio policy pays a fixed percentage of earnings, and the dividend will rise when earnings increase.

31 Multiple Choice 1 / 1 point

If the analysts' beliefs about the increase in sales are correct, the change in dividend policy that Avignon would *most likely* make would be to:

☒ declare a special dividend.

Correct Answer Feedback: Correct. If analysts are correct that the change in sales is temporary, the company is most likely to declare a special dividend. Companies, particularly in cyclical industries, may choose to use special dividends to distribute more earnings during strong earning years.

☐ increase the quarterly dividend amount.

Incorrect Answer Feedback: Incorrect. Most companies strive to maintain or increase their dividends and will not increase the regular dividend unless they believe they can continue to pay at or above that level. Rather, they will pay an extra dividend at the end of the year when earnings are unusually good.

- ☐ cut the quarterly dividend in anticipation of next year's sales forecast.

Incorrect Answer Feedback: Incorrect. A record of consistent or increasing dividends is widely interpreted as a signal of profitability, and most companies strive to not reduce dividends.

32 Multiple Choice 1 / 1 point

Which of Lee's statements to Carlyle about share repurchases is *least* accurate?

- ☒ Statement 1

Correct Answer Feedback: Correct. Statement 1 is least accurate. Although the amount of cash distributed is the same, only those shareholders who have their shares purchased by the company will receive any cash.

- ☐ Statement 2

Incorrect Answer Feedback: Incorrect. Statement 2 is correct. Management is not obligated to follow through after stating an intention to repurchase shares, whereas they must meet a declared dividend statement.

- ☐ Statement 3

Incorrect Answer Feedback: Incorrect. Statement 3 is correct. Research showed that 45% of private repurchases between 1984 and 2001 were actually made at discounts, indicating that many direct negotiation repurchases are generated by the liquidity needs of large investors who are in a weak negotiating position.

33 Multiple Choice 0 / 1 point

If SpeedyPro had used all of its surplus cash to repurchase its shares, based on Exhibit 2, the percentage increase in EPS would have been *closest* to:

- ☐ 10%.

Incorrect Answer Feedback: Incorrect because it applies the price premium (10%) to the EPS.

- ☐ 25%.

Correct Answer Feedback: Correct. The increase in EPS was 25%, calculated as follows:

Steps	Calculation		EPS
EPS before the repurchase	from Exhibit 2	---	\$1.24
Surplus cash available for repurchase	---	\$836 million	---
# share outstanding before repurchase	---	100 million	---
Share price at time of repurchase	---	\$38.00	---
Price premium for repurchase	---	10%	---
Repurchase price per share	$\$38.00 \times 110\%$	\$41.80	---
# shares repurchased	$\$836 \text{ million} \div \41.80 per share	20 million	---
# shares outstanding after repurchase	$100 \text{ million} - 20 \text{ million}$	80 million	---
EPS after the repurchase	$\$124 \text{ million} \div 80 \text{ million shares}$	---	\$1.55
% increase in EPS	$(\$1.55 - \$1.24) \div \$1.24$	---	25%

☐ 28%.

Incorrect Answer Feedback: Incorrect because it fails to include share premium.

- Shares to purchase (in millions) = $836/38 = 22$.
- Shares outstanding after repurchase (in millions) = $100 - 22 = 78$.
- EPS after the repurchase = $\$124/78 = \1.59 .
- Change in EPS = $1.59/1.24 = 128\% = 28.00\%$ increase.

34 Multiple Choice 1 / 1 point

Based on Exhibit 2, SpeedyPro *most likely* repurchased shares using:

☐ open market purchases.

Incorrect Answer Feedback: Incorrect. Open market purchases are market based and can be timed to avoid price impact.

☒ a fixed-price tender offer.

Correct Answer Feedback: Correct. SpeedyPro offered a 10% premium over the current price. This is most consistent with a fixed-price tender offer, which normally requires a premium. Negotiated purchase agreements are almost as likely to take place at prices lower than market because they are at prices above market, particularly when shareholders are trying to meet liquidity needs. Open market purchases are market based and can be timed to avoid price impact.

☐ a negotiated purchase agreement.

Incorrect Answer Feedback: Incorrect. Negotiated purchase agreements are almost as likely to take place at prices lower than market as they are at prices above market when shareholders are trying to meet liquidity needs.

35 Multiple Choice 0 / 1 point

The *best* answer to Carlyle's question about the potential violation of the debt covenants is that the covenant:

- ☐ is not violated if Avignon repurchases shares.

Incorrect Answer Feedback: Incorrect. The D/E is above the 35% threshold when the repurchase is financed with debt.



- ☒ will be violated if Avignon uses debt to finance the repurchase.

Correct Answer Feedback: Correct. The debt-financed repurchase increases the debt-to-equity ratio above the 35% threshold and thus violates the debt covenant.

Book value of equity (millions)	Exhibit 3	C\$3,600
Cash available for purchase (millions)	Exhibit 3	C\$155
Debt-to-equity ratio (D/E) before	Exhibit 3	30.0%
Book value of debt (millions)	$30\% \times \text{C\$}3,600$	C\$1,080
D/E with cash repurchase	$\text{C\$}1,080 \div (\text{C\$}3,600 - \text{C\$}155)$	31.3%
D/E with debt-financed repurchase	$(\text{C\$}1,080 + \text{C\$}155) \div (\text{C\$}3,600 - \text{C\$}155)$	35.8%



- ☐ will be violated if Avignon uses the surplus cash to finance the repurchase.

Incorrect Answer Feedback: Incorrect. The D/E stays below 35% when the company uses the surplus cash to finance the repurchase.

Vignette

Michael Thornhill is CFO at American Automotive Parts & Service Corporation (AAPS), an automotive manufacturer that specializes in engine components in a number of established vehicle production lines. He is preparing for a meeting with members of the executive committee to discuss and prepare recommendations for the board to determine if AAPS should make any changes to its dividend policy and structure over the next fiscal year.

In preparation for this meeting, Thornhill summarizes AAPS's historical dividends and related information in Exhibit 1.

Exhibit 1

Dividend frequency	Quarterly
Dividend amount, current	\$0.50/share
Number of shares outstanding	10,000,000
Target payout ratio	20%
Corporate tax rate	35%
Share price – 52-week low-high	\$7.84 – \$10.62
Share price – closing	\$9.89

In the meeting, the first topic discussed is the recently completed sale of AAPS's brake parts division. The sale generated \$29 million in cash proceeds. Thornhill recommends that the company issue a one-time dividend of \$0.95/share to existing shareholders.

The Chief Operating Officer (COO) highlights that the company has a number of potential investments options and suggests that the company consider a 4% stock dividend instead. Thornhill agrees to prepare a summary of the effect the stock dividend would have on the company's results assuming a purchase price per share based on the current market price.

The following statements about the stock dividend are discussed:

Statement The stock dividend would be negatively received by shareholders because it would
1: increase their tax liability.

Statement The stock dividend would have a more adverse effect on the company's financial
2: leverage ratios which are being closely monitored by the bank.

Statement The flotation costs of the stock dividend would be lower than a traditional issuance
3: and should be considered in their analysis.

As a result of the extended discussion and different perspectives on dividend payout structures, Thornhill suggests an alternative approach could be a share buyback transaction. The following statements are made:

Statement AAPS should negotiate a greenmail transaction directly with a major activist
1: shareholder in order to secure a discount to the current market price.

Statement If there is a concern about the perception of excluding other shareholders with the
2: greenmail transaction, a fixed price tender offer could be accomplished quickly and allow them to purchase a pro rata portion of shares from all interested shareholders.

Statement AAPS should announce an open market repurchase program as it would provide
3: flexibility to cancel the share purchases if the company is able to identify another project that would be a more effective use of funds.

36 Multiple Choice 0 / 1 point

The one-time dividend Thornhill recommends is *best* described as a:

☐ scrip dividend.

Incorrect Answer Feedback: Incorrect because with a stock dividend (also known as a bonus issue of shares or a scrip dividend), the company distributes additional shares of its common stock to shareholders instead of cash. AAPS sold a division and distributed a portion of the proceeds back to shareholders and the one-time \$0.95/share dividend is a liquidating dividend.

☒ special dividend.

Incorrect Answer Feedback: Incorrect because a special dividend is either a dividend paid by a company that does not pay dividends on a regular schedule or a dividend that supplements regular cash dividends with an extra payment. AAPS pays dividends and the one-time \$0.95/share dividend is in relation to the brake parts division which is a liquidating dividend.

☐ liquidating dividend.

Correct Answer Feedback: Correct because a liquidating dividend arises when a company sells a portion of its business for cash and the proceeds are distributed to shareholders. In this case, AAPS sold a division and distributed a portion of the proceeds back to shareholders and the one-time \$0.95/share dividend is a liquidating dividend.

37 Multiple Choice 0 / 1 point

If AAPS follows the COO's suggestion, the expected share price after the stock dividend is *closest* to:

☒ \$9.49.

×

Incorrect Answer Feedback: Incorrect because it assumes the 4% stock dividend reduces the stock price by 4%

$$= \$9.89 \times (1 - 0.04) = \$9.49$$

☐ \$9.51.

✓

Correct Answer Feedback: Correct. For a stock dividend, the total market value of the company is unaffected by the stock dividend because the decrease in the share price is exactly offset by the increase in the number of shares outstanding. The total cost basis of the shares remains the same; however, the number of shares owned increases.

Assuming a shareholder owns 100 shares.

$$\text{Total value} = \$989.00 [\$9.89 \times 100]$$

With a 4.0% stock dividend, the number of shares would increase to 104.

The new share price = Total value / new number of shares

$$= \$989.00 / 104$$

$$= \$9.51$$

☐ \$10.29.

Incorrect Answer Feedback: Incorrect because it multiplies the current share price with the expected stock dividend growth

$$= \$9.89 \times (1 + 0.04)$$

$$= \$10.29$$

38 Multiple Choice 1 / 1 point

Which of the following statements about the effect of the stock dividend is *most* accurate?

☐ Statement 1

Incorrect Answer Feedback: Incorrect because a stock dividend is generally not taxable to shareholders because a stock dividend merely divides the "pie" (the market value of shareholders' equity) into smaller pieces.

☐ Statement 2

Incorrect Answer Feedback: Incorrect because a stock dividend has no economic impact on a company. Stock dividends do not affect assets or shareholders' equity. Although retained earnings are reduced by the value of the stock dividends paid (i.e., by the number of shares issued $\times$ price per share), contributed capital increases by the same amount (i.e., the value of the shares issued). As a result, total shareholders' equity does not change. Neither stock dividends nor stock splits affect liquidity ratios or financial leverage ratios.

☒ Statement 3

✓

Correct Answer Feedback: Correct. Flotation costs are the costs associated with selling shares to the public that would include underwriters' fees, legal costs, registration expenses, and possible negative price effects and would be lower with a direct stock dividend to existing owners. The company did not have to spend any actual money issuing a dividend.

Multiple Choice 0 / 1 point

In the share buyback discussion, the statement that is the *least* accurate is:

☐ Statement 1.

Correct Answer Feedback: Correct because in a greenmail transaction, the company buys back the would-be suitor's shares at a premium to the market price, not at a discount. Unsuccessful takeover attempts have ended with the company buying back the would-be suitor's shares at a premium to the market price, referred to as a greenmail transaction, often to the detriment of remaining shareholders.

☒ Statement 2.

Incorrect Answer Feedback: Incorrect because it is an accurate statement. A fixed price tender offer can be extended to all shareholders and can set at a fixed date in the near future. A company will make a fixed price tender offer to repurchase a specific number of shares at a fixed price that is typically at a premium to the current market price. By setting a fixed date, such as 30 days in the future, a fixed price tender offer can be accomplished quickly.

☐ Statement 3.

Incorrect Answer Feedback: Incorrect because it is an accurate statement. The open market share repurchase method gives the company maximum flexibility. Open market repurchases are the most flexible option for a company because there is no legal obligation to undertake or complete the repurchase program; a company may not follow through with an announced program for various reasons, such as unexpected cash needs for liquidity, acquisitions, or capital expenditures.

Vignette

Andrea Larsen is the CFO of Fjord Marine Industries (FMI), a Norwegian pleasure boat manufacturer. Larsen is reviewing FMI's dividend policy in advance of recommending a course of action to the Board of Directors regarding the current year dividend payment.

FMI's business is influenced by many factors, including commodity prices, interest rates and general economic

Exhibit 1

	Current Year	Prior Year	Two Years Ago
Pre-tax income (in NOK millions)	600	450	400
After-tax net income (in NOK millions)	420	315	300
Dividends paid (in NOK millions)	---	120	115
Number of shares outstanding (in millions)	6	6	6

FMI's shares are currently trading at NOK 460.

Exhibit 1 shows that FMI had a very strong year in the current year, generating an additional NOK 105 million net income over the previous year. Larsen considers four possible options in addition to the regular dividend to return a portion of this additional capital to shareholders in the current year:

Option 1: Issue a stock dividend of 20%.

Option 2: Increase the annual per share dividend by 20% more than the prior year.

Option 3: Maintain the current annual dividend and pay a special dividend of NOK 10 per share.

Option 4: Repurchase NOK 200 million worth of shares at a NOK 40 premium to the current share price.

40 Multiple Choice 1 / 1 point

For FMI, Option 1 would *most likely*:

☐ affect its capital structure.

Incorrect Answer Feedback: Incorrect because from a company's perspective, the key difference between a stock dividend and a cash dividend is that a cash dividend affects a company's capital structure, whereas a stock dividend has no economic impact on a company.



☒ reduce the volatility of its share price.

Correct Answer Feedback: Correct because the payment of a stock dividend also helps increase the stock's float, which improves the liquidity of the shares and dampens share price volatility.

☐ increase the value of each shareholder's ownership position in the company.

Incorrect Answer Feedback: Incorrect because a stock dividend does not change the value of each shareholder's ownership position because the increase in the number of shares held is accompanied by an offsetting decrease in earnings per share, and other measures of value per share, resulting from the greater number of shares outstanding.

41 Multiple Choice 0 / 1 point

Compared to option 2, option 3 results in a total current year dividend payment per share that is:



☐ lower by NOK 6.

Incorrect Answer Feedback: Incorrect because under option 2, the current year dividend payment would be $(\text{NOK } 120 \text{ million} \times 1.2) / 6 \text{ million shares} = \text{NOK } 24 \text{ per share}$. Under option 3 the total dividend paid would be $\text{NOK } 120 \text{ million} / 6 \text{ million} + \text{NOK } 10 = \text{NOK } 30$. This response reverses the two options: $\text{NOK } 24 - \text{NOK } 30 = -\text{NOK } 6$.



☒ higher by NOK 6.

Correct Answer Feedback: Correct because under option 2, the current year dividend payment would be $(\text{NOK } 120 \text{ million} \times 1.2) / 6 \text{ million shares} = \text{NOK } 24 \text{ per share}$. Under option 3 the total dividend paid would be $\text{NOK } 120 \text{ million} / 6 \text{ million} + \text{NOK } 10 = \text{NOK } 30$. The per share dividend increase is therefore $\text{NOK } 30 - \text{NOK } 24 = \text{NOK } 6$. An extra dividend or special dividend (also known as an irregular dividend) is either a dividend paid by a company that does not pay dividends on a regular schedule or a dividend that supplements regular cash dividends with an extra payment.

☐ higher by NOK 7.

Incorrect Answer Feedback: Incorrect because under option 2, the current year dividend payment would be (NOK 120 million $\times$ 1.2) / 6 million shares = NOK 24 per share. Under option 3 the total dividend paid would be NOK 120 million / 6 million + NOK 10 = NOK 20 + NOK 10 = NOK 30. This response incorrectly uses the dividend payment from two years ago: NOK 115 million / 6 million shares $\approx$ NOK 19.17 per share. Therefore NOK 30 – (NOK 19.17 $\times$ 1.2) = NOK 30 – NOK 23 = NOK 7.

42 Multiple Choice 1 / 1 point

Based on Exhibit 1, the safety of FMI's dividend in the prior year compared to two years ago:

☐ decreased.

Incorrect Answer Feedback: Incorrect because the dividend coverage ratio (after-tax net income / dividends paid) increased from $300/115 \approx 2.609$ from two years ago to $315/120 = 2.625$ in the prior year, which means the probability of receiving a dividend increased making the dividend more safe.

☐ remained the same.

Incorrect Answer Feedback: Incorrect because the dividend coverage ratio (after-tax net income / dividends paid) increased from $300/115 \approx 2.609$ from two years ago to $315/120 = 2.625$ in the prior year, which means the probability of receiving a dividend increased making the dividend more safe.

☒ increased.

Correct Answer Feedback: Correct because the dividend coverage ratio (after-tax net income / dividends paid) increased from $300/115 \approx 2.609$ from two years ago to $315/120 = 2.625$ in the prior year, which means the probability of receiving a dividend increased making the dividend more safe.

43 Multiple Choice 1 / 1 point

If option 4 is implemented using surplus cash, which of the following is *most likely* to decrease? FMI's:

☒ total book value.

Correct Answer Feedback: Correct because share repurchases involve a reduction in shareholder equity and therefore a reduction in the total book value reported on the balance sheet. Whether the book value per share increases or decreases depends on whether the market price per share is above or below the book value per share.

☐ debt-to-equity ratio.

Incorrect Answer Feedback: Incorrect because share repurchases affect both the balance sheet and income statement. Both assets and shareholders' equity decline if the repurchase is made with surplus cash. As a result, leverage increases. Debt ratios (leverage) will increase even more if the repurchase is financed with debt. The debt-to-equity ratio would therefore increase following a cash share repurchase.

☐ debt-to-assets ratio.

Incorrect Answer Feedback: Incorrect because share repurchases affect both the balance sheet and income statement. Both assets and shareholders' equity decline if the repurchase is made with surplus cash. As a result, leverage increases. Debt ratios (leverage) will increase even more if the repurchase is financed with debt. The debt-to-assets ratio would therefore increase following a cash share repurchase as the denominator (total assets) decreases.

